

MM-Mixer: Axis-Wise Fusion of Utterance-Level Features for Multimodal Emotion Recognition in Conversations

Anonymous submission

Abstract

Multimodal Emotion Recognition in Conversation (MERC) predicts utterance-level emotions from textual, acoustic, and visual evidence. We study fusion when each modality is represented by a pre-extracted utterance vector. MM-Mixer applies modality-specific gating and modality-centric cross-modal alignment, then projects each aligned vector into a fixed set of input-dependent latent coordinates. Its Axis-wise Multimodal Mixer combines coordinates within each modality, exchanges information across modalities, and transforms feature channels. A query-based main fusion path is supplemented by Explicit Pairwise Interaction Residual Correction (EPIRC), which adds low-rank products of the three modality pairs. In the reported IEMOCAP and MELD comparisons, MM-Mixer has the highest mean overall accuracy and weighted F1 among the listed models; the tabulated downstream parameter counts are lower than those of recent fusion baselines. These results characterize fusion of pre-extracted features, while the available aggregate ablations do not establish seed-stable effects of individual components.

Introduction

Multimodal Emotion Recognition in Conversation (MERC) aims to identify the emotion expressed by each utterance in a dialogue by integrating textual, acoustic, and visual signals. Early ERC studies mainly focused on text-based conversational context and speaker dependencies through recurrent state tracking and graph-based reasoning (Majumder et al. 2019; Ghosal et al. 2019). With the increasing availability of multimodal conversational data, subsequent MERC methods have incorporated acoustic and visual signals through multimodal graph learning (Hu et al. 2021), cross-modal attention (Mao et al. 2021), and more expressive fusion mechanisms (Ma et al. 2024; Lyu et al. 2026). These approaches have substantially advanced multimodal representation learning for MERC by enabling information exchange among heterogeneous modality streams.

Figure 1 provides concrete examples from MELD (Poria et al. 2019): the same speaker uses the exact same current words with different annotated emotions. These examples

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illustrate the task setting; matching current words does not imply matching the history-aware text features used by the model.

In the setting of pre-extracted utterance-level features, text, audio, and vision each enter downstream fusion as a single vector. This is a compact input representation, but information discarded by the feature extractors cannot be recovered by the fusion model. We therefore examine how to organize interactions among learned coordinates of these vectors, without interpreting those coordinates as temporal observations or predefined emotion cues.

MM-Mixer first uses Adaptive Gating to calibrate each modality with a shared reference and Modality-centric Cross-modal Alignment to retrieve information from the other modalities. It then applies one shared projection to each aligned modality vector and reshapes the output into a fixed-length latent sequence. Each element is an input-dependent feature coordinate group; the sequence does not index dialogue turns, acoustic frames, or visual frames, and its elements have no assigned semantic meaning. The Axis-wise Multimodal Mixer combines these coordinates within a modality, routes information across modalities, and transforms feature channels. Its distinction from a flat feature transformation is the chosen axis-specific operations and parameter sharing, not an established increase in representational power.

Query-based integration forms the main fused representation. EPIRC adds a parallel, zero-initialized residual computed from low-rank products of the three modality pairs. This explicitly parameterizes pairwise multiplicative terms alongside the main path; whether they improve a particular emotion class or contribute independently must be assessed from matched experiments.

The reported IEMOCAP (Busso et al. 2008) and MELD (Poria et al. 2019) comparisons use history-only inputs and show strong overall accuracy and weighted F1 among the listed methods. Text remains the strongest single modality in the reported modality ablation, and all three inputs are pre-extracted utterance features. The results therefore support MM-Mixer as a compact downstream fusion design in this setting, while baseline differences and aggregate ablations alone do not identify the cause or stability of each component’s apparent contribution.

Our main contributions are summarized as follows:

Same words, different annotated emotions

Same speaker: Ross | Same current utterance: "Hey!"

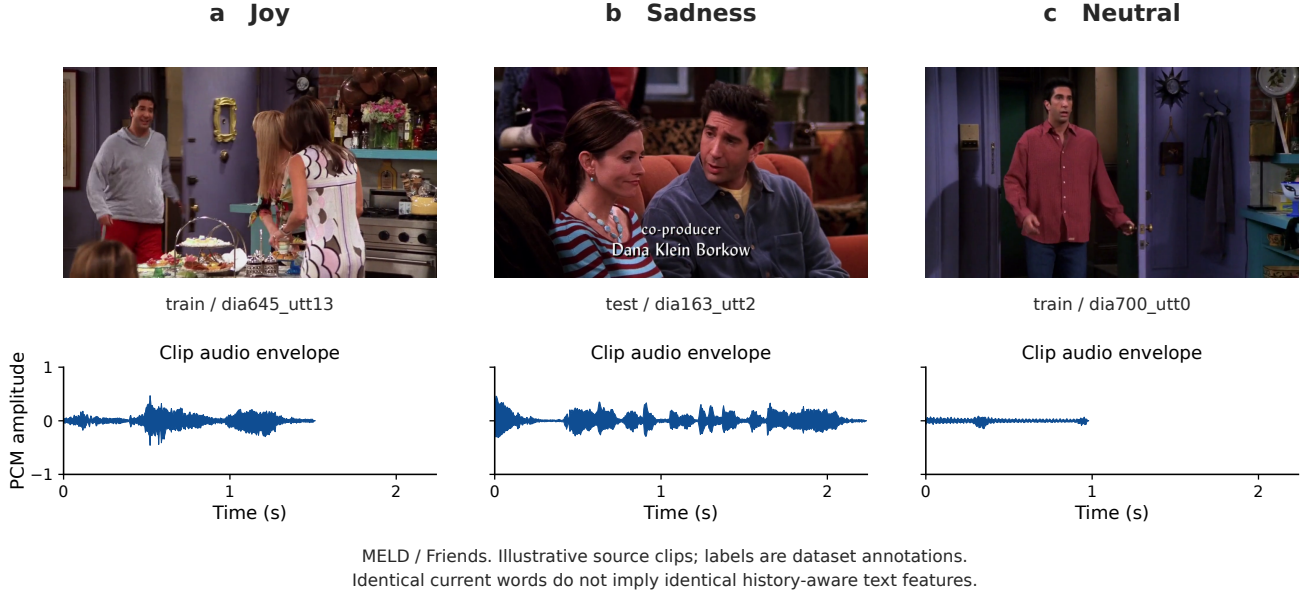


Figure 1: Same current words, different emotion labels in MELD. Ross’s “Hey!” is labeled joy (train/dia645_utt13), sadness (test/dia163_utt2), and neutral (train/dia700_utt0). Panels show original video frames and clip-audio envelopes on common time and PCM amplitude scales. Selection matches the current words and speaker, seeks distinct labels, and checks speaker visibility. Identical current words do not imply identical history-aware text features. These are qualitative task illustrations without model-prediction comparisons.

- We propose MM-Mixer to organize interactions among pre-extracted utterance-level modality representations for MERC.
- We introduce an Axis-wise Multimodal Mixer that applies distinct operations along input-dependent latent coordinates, modalities, and hidden feature channels.
- We introduce EPIRC as a low-rank pairwise multiplicative residual to the main fusion path.
- We report IEMOCAP and MELD comparisons and descriptive component and modality ablations under the stated evaluation setting.

Related Work

Context Modeling in ERC. Conversational context provides important evidence for interpreting an utterance whose emotion cannot be determined in isolation. Sequence-based methods model this evidence by tracking dialogue and speaker states. For example, DialogueRNN (Majumder et al. 2019) maintains speaker-specific states together with a global dialogue state. Graph-based methods instead represent utterances as nodes and propagate information through speaker-dependent relations, as in DialogueGCN (Ghosal et al. 2019). DAG-ERC (Shen et al. 2021) further organizes information flow with a directed acyclic graph that combines local and long-range context. These methods mainly differ in how

information is propagated across utterances and speakers. More recently, PRC-Emo (Li et al. 2026) combines emotion-sensitive prompting, demonstration retrieval, and a curriculum based on weighted emotion shifts between same- and different-speaker utterances to model explicit and implicit contextual cues.

MM-Mixer does not introduce a new dialogue topology and focuses on multimodal interaction after the causal textual context has been encoded.

Multimodal Interaction and Fusion. MERC methods must combine heterogeneous textual, acoustic, and visual signals while retaining information specific to each modality. MMGCN (Hu et al. 2021) constructs a multimodal graph to model cross-modal and speaker-dependent relations, while MMDAG (Xu et al. 2022) extends directed acyclic context modeling to multimodal inputs. Transformer-based approaches use intra- and inter-modal attention for direct information exchange. DialogueTRM (Mao et al. 2021) follows this strategy to model multimodal emotional dynamics, and SDT (Ma et al. 2024) combines intra- and inter-modal Transformers with gated fusion and unimodal self-distillation. Other methods introduce task-specific interaction structures. ECERC (Zhang and Tan 2025) separates target-utterance evidence from contextual causes, whereas GS-MCC (Ai et al. 2025) models multimodal conversations through graph-spectral operators. Recent work has

also moved toward asymmetric or component-aware fusion. EMART (Seong et al. 2026) treats text as the primary information source and audio as a complementary signal, coupling modality-aware fusion with an emotion-wheel-guided supervised contrastive objective. DnR (Mai, Nguyen, and Le 2026) explicitly decomposes multimodal representations into unique, pairwise redundant, and synergistic components before applying component-specific refinement. Together, these methods organize multimodal reasoning through graph propagation, attention-based exchange, or task-specific interaction structures.

Compared with these methods, MM-Mixer projects each utterance-level modality vector into a structured tensor and assigns separate transformations to its learned-coordinate, modality, and hidden-feature axes. These axes are defined by the model parameterization; the learned coordinates are not observed temporal tokens or semantically labeled parts.

Axis-Factorized and Multiplicative Interaction. Axis-factorized networks assign different transformations to different dimensions of a representation. MLP-Mixer (Tolstikhin et al. 2021) separates token mixing from channel mixing, demonstrating that interactions along these two axes can be modeled by dedicated MLPs. Multiplicative fusion provides another way to expose cross-modal dependencies. Factorized bilinear pooling and low-rank multimodal fusion approximate tensor interactions without constructing a full outer-product tensor (Yu et al. 2017; Liu et al. 2018). In MERC, CSS (Lyu et al. 2026) uses low-rank polynomial fusion as its main multimodal fusion operator and coordinates multiple objectives through Pareto gradient modulation.

Methodology

MM-Mixer Overview

Given a dialogue $\{u_t\}_{t=1}^N$ containing N utterances, MERC aims to predict an emotion label $y_t \in \{1, \dots, C\}$ for each utterance u_t . The model input contains a textual feature $\mathbf{x}_t^T \in \mathbb{R}^{d_T}$, an acoustic feature $\mathbf{x}_t^A \in \mathbb{R}^{d_A}$, and a visual feature $\mathbf{x}_t^V \in \mathbb{R}^{d_V}$. The superscripts T , A , and V identify the textual, acoustic, and visual modalities, respectively. The textual feature represents the target utterance together with its preceding dialogue context, whereas the acoustic and visual features describe the target utterance. All inputs exclude utterances after u_t . MM-Mixer learns a mapping f_θ from the three modality features to the emotion space and selects the class with the highest predicted probability:

$$\hat{y}_t = \underset{c \in \{1, \dots, C\}}{\operatorname{argmax}} f_\theta(\mathbf{x}_t^T, \mathbf{x}_t^A, \mathbf{x}_t^V)_c \quad (1)$$

where f_θ denotes the model parameterized by θ , and $\hat{y}_t$ is the predicted emotion label of u_t . For clarity, the utterance index t is omitted below.

As illustrated in Figure 2, MM-Mixer first employs pre-trained modality-specific encoders to extract utterance-level unimodal features and projects them into a common feature space. Modality-specific feature gates suppress irrelevant channels. Adaptive Gating (AG) then constructs a shared multimodal reference and uses it to calibrate each modality, after which Modality-centric Cross-modal Alignment (MCA) uses each modality to query complementary

information from the other two modalities. The aligned representations are then organized along the learned-sequence, modality, and hidden-feature axes by the Axis-wise Multimodal Mixer (AMM). Finally, query-based integration forms the main fusion pathway, while the Explicit Pairwise Interaction Residual Correction (EPIRC) branch provides an additive low-rank multiplicative correction. The fused prediction and three unimodal predictions are jointly optimized during training.

Unimodal Feature Extraction

MM-Mixer first deploys pre-trained modality-specific encoders to extract utterance-level unimodal features $\mathbf{x}^T$, $\mathbf{x}^A$, and $\mathbf{x}^V$. Since the input modalities have different feature dimensions and feature distributions, each modality is projected into a common feature space to reduce their heterogeneity, followed by filtering irrelevant information with a feature gating as follows:

$$\begin{aligned} \mathbf{e}^m &= \text{Dropout}(\text{GELU}(\text{LN}(\mathbf{W}_p^m \mathbf{x}^m + \mathbf{b}_p^m))) \\ \text{coff}^m &= \sigma(\mathbf{W}_{g2}^m \text{ReLU}(\text{LN}(\mathbf{W}_{g1}^m \mathbf{e}^m + \mathbf{b}_{g1}^m)) + \mathbf{b}_{g2}^m) \\ \mathbf{u}^m &= \mathbf{e}^m \odot \text{coff}^m \end{aligned} \quad (2)$$

where $\mathbf{e}^m \in \mathbb{R}^D$ is the unimodal representation, $\mathbf{u}^m \in \mathbb{R}^D$ is its refined version, and $m \in M$, with $M = \{T, A, V\}$. $\mathbf{W}_p^m$, $\mathbf{W}_{g1}^m$ and $\mathbf{W}_{g2}^m$ are learnable weights. $\mathbf{b}_p^m$, $\mathbf{b}_{g1}^m$ and $\mathbf{b}_{g2}^m$ are learnable biases. Dropout, GELU, ReLU, and LN denote the dropout, GELU activation, RELU activation, and layer normalization, respectively. σ denotes the Sigmoid function. Here, it is used to suppress irrelevant channels while preserving one feature vector per modality.

Adaptive Gating

Given the filtered unimodal features, the AG module constructs a shared multimodal reference as follows:

$$\begin{aligned} \alpha &= \text{softmax}(\text{MLP}([\mathbf{u}^T; \mathbf{u}^A; \mathbf{u}^V])) \\ \mathbf{c} &= \sum_{m \in M} \alpha_m \mathbf{u}^m \end{aligned} \quad (3)$$

where $\alpha \in \mathbb{R}^3$ contains the normalized modality coefficients, and $\mathbf{c} \in \mathbb{R}^D$ is their weighted multimodal reference. The shared reference then conditions a separate channel gate to calibrate modality features for each modality:

$$\mathbf{f}^m = [\sigma(\text{MLP}_c^m([\mathbf{u}^m; \mathbf{c}])) + \epsilon_g] \odot \mathbf{u}^m \quad (4)$$

Here, $\mathbf{f}^m \in \mathbb{R}^D$ is the calibrated modality feature. The context-conditioned sigmoid coefficients perform channel selection, while ϵ_g preserves a direct contribution from $\mathbf{u}^m$.

Modality-centric Cross-modal Alignment

The MCA module comprises three directional cross-attention branches. For each $m \in M$, one branch uses the calibrated feature of modality m as its query, while the other two modalities provide the keys and values:

$$\begin{aligned} \mathbf{K}^m &= \mathbf{V}^m = \text{Stack}\{\mathbf{f}^n \mid n \in M, n \neq m\} \\ \mathbf{h}^m &= \text{MHA}^m(\mathbf{f}^m, \mathbf{K}^m, \mathbf{V}^m) + \epsilon_a \mathbf{f}^m \end{aligned} \quad (5)$$

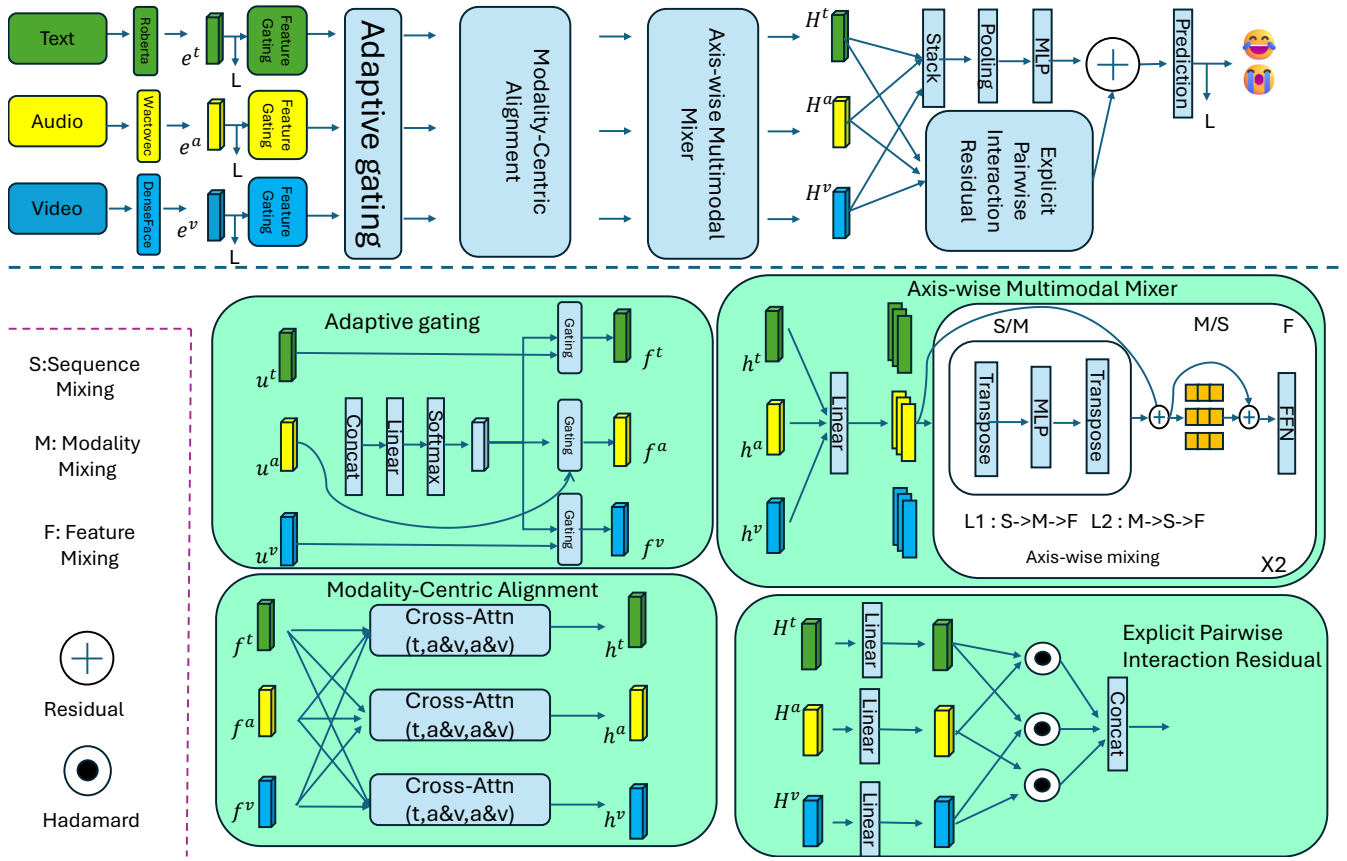


Figure 2: Overview of MM-Mixer. Pre-extracted textual, acoustic, and visual features are projected and gated before adaptive refinement and modality-centric cross-modal alignment. Two alternating axis-wise mixer blocks organize the resulting representations along the latent-sequence, modality, and feature dimensions. Query-based integration forms the main prediction pathway, while explicit low-rank pairwise interactions provide a residual correction. Layer normalization and internal residual connections are omitted for clarity.

Here, $\mathbf{K}^m$ and $\mathbf{V}^m \in \mathbb{R}^{2 \times D}$ contain the two peer-modality vectors. MHA^m is the multi-head attention layer for query modality m , and ϵ_a controls the query residual. The output $\mathbf{h}^m \in \mathbb{R}^D$ is the cross-modal aligned feature of modality m .

Axis-wise Multimodal Mixer

AMM organizes the cross-modal aligned features $\mathbf{h}^m$ along three model-defined axes: projected coordinates within a modality, modality identity, and hidden feature channels. Separate operators specify which coordinates may interact at each stage. Sequence and feature mixing transform representations within a modality; modality mixing is the operator that directly exchanges information across modalities.

Figure 3 illustrates the learned projections and the operations along the three representation axes.

Sequence construction. A shared affine map transforms each D -dimensional vector $\mathbf{h}^m$ into SD values, which are reshaped into S input-dependent elements of width D . Stacking the three outputs gives $\mathbf{Z}^0 \in \mathbb{R}^{3 \times S \times D}$:

$$\mathbf{Z}^0 = \text{Stack}_{m \in M} [\text{reshape}_{S \times D} (\mathbf{W}_s \mathbf{h}^m + \mathbf{b}_s)] \quad (6)$$

where the parameters $\mathbf{W}_s$ and $\mathbf{b}_s$ are shared by T , A , and

V , and the modality axis follows the fixed order V, A, T . “Learned elements” refers to the projected values produced for each input, not to an independent bank of trainable tokens. Projection and reshaping are themselves a feature transformation. The subsequent axis-specific operators impose an interaction pattern and shared parameters across coordinates; the elements have no temporal order or guaranteed semantic interpretation.

For the remainder of this subsection, $[\cdot]_{m,s,d}$ denotes an entry at modality m , projected coordinate s , and feature channel d .

Axis-wise mixing. Each mixer block contains three pre-normalized residual operators. Let $\mathbf{Z} \in \mathbb{R}^{3 \times S \times D}$ denote a generic input to an axis-mixing operator. Sequence mixing applies an MLP along the S axis while holding the modality and feature indices fixed:

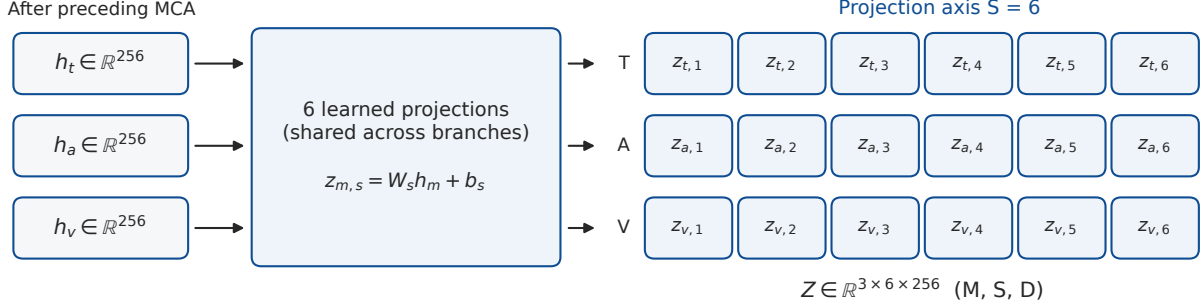
$$[\mathcal{S}(\mathbf{Z})]_{m,:,d} = [\mathbf{Z}]_{m,:,d} + \text{MLP}_S([\text{LN}_S(\mathbf{Z})]_{m,:,d}). \quad (7)$$

Here, MLP_S maps the sequence axis through a hidden layer and returns an output of the same width.

Modality mixing instead holds (s, d) fixed and transforms

AMM: learned projection views and axis-wise interaction

a Expand each branch representation



b Mix axes, then aggregate

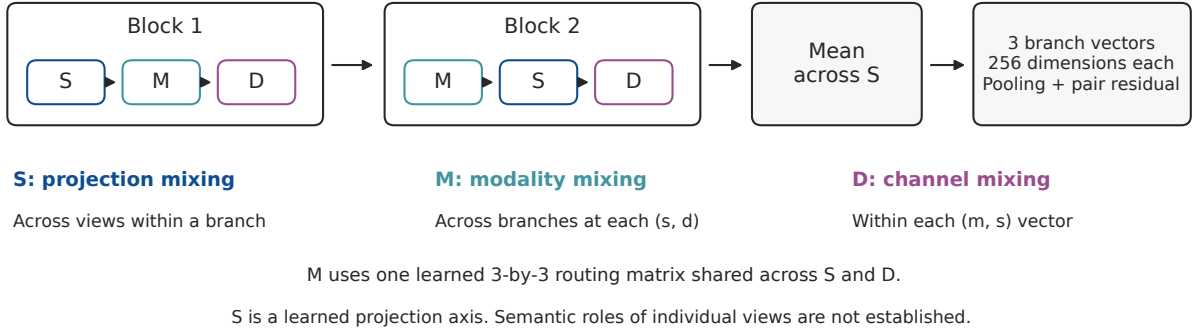


Figure 3: Learned projection views and axis-wise interaction in AMM. Each modality-branch representation after MCA is mapped to six learned projections shared across branches. Two blocks use projection–modality–channel and modality–projection–channel update orders, respectively, before averaging over the projection axis. Modality routing uses one learned 3-by-3 matrix shared across projection and channel positions. Layer normalization and residual additions are omitted for clarity. Individual views have no predefined semantic roles.

the three-vector $[[Z]_{V,s,d}; [Z]_{A,s,d}; [Z]_{T,s,d}]$:

$$[\mathcal{M}(\mathbf{Z})]_{:,s,d} = [\mathbf{Z}]_{:,s,d} + \mathbf{W}_r [\text{LN}_M(\mathbf{Z})]_{:,s,d}. \quad (8)$$

The bias-free routing matrix $\mathbf{W}_r \in \mathbb{R}^{3 \times 3}$ is shared across all sequence coordinates and feature channels, providing a common cross-modal transformation at every sequence-feature coordinate.

Feature mixing completes the block by applying a channel-wise MLP to every modality–sequence pair:

$$[\mathcal{F}(\mathbf{Z})]_{m,s,:} = [\mathbf{Z}]_{m,s,:} + \text{MLP}_D([\text{LN}_F(\mathbf{Z})]_{m,s,:}). \quad (9)$$

Here, $\mathcal{S}$, $\mathcal{M}$, and $\mathcal{F}$ denote sequence, modality, and feature mixing, respectively. The feature MLP follows $D \rightarrow D_h \rightarrow D$ with GELU, where D_h is its hidden width.

We stack two mixer blocks with different operation orders:

$$\begin{aligned} \mathbf{Z}^1 &= (\mathcal{F}^1 \circ \mathcal{M}^1 \circ \mathcal{S}^1)(\mathbf{Z}^0) \\ \mathbf{Z}^2 &= (\mathcal{F}^2 \circ \mathcal{S}^2 \circ \mathcal{M}^2)(\mathbf{Z}^1). \end{aligned} \quad (10)$$

$\mathbf{Z}^\ell$ denotes the output tensor of mixer block ℓ , and the composition symbol $\circ$ is evaluated from right to left. Therefore, the

first mixer block organizes sequence elements before routing across modalities, whereas the second routes modalities before sequence refinement. Averaging over the sequence axis returns one vector per modality:

$$\mathbf{H}^m = \frac{1}{S} \sum_{s=1}^S \mathbf{Z}_{m,s,:}^2 \quad (11)$$

$\mathbf{H}^m \in \mathbb{R}^D$ is the mixer output for modality m .

Multimodal Fusion with Explicit Pairwise Interaction Residual Correction

Given the three modality representations produced by AMM, the multimodal fusion contains a main fusion path and a parallel residual path. The main path first stacks the mixer outputs along the modality axis:

$$\mathbf{H} = \text{Stack}(\mathbf{H}^V, \mathbf{H}^A, \mathbf{H}^T) \quad (12)$$

where $\mathbf{H} \in \mathbb{R}^{3 \times D}$. Three learnable queries then aggregate the stacked modality representations via multi-head attention:

$$\mathbf{P} = \text{MHA}(\mathbf{Q}_p, \mathbf{H}, \mathbf{H}) \quad (13)$$

Here, $\mathbf{Q}_p \in \mathbb{R}^{3 \times D}$ contains three learned query vectors, and $\mathbf{P}$ contains their pooled outputs. The outputs are flattened and integrated to form the main fused representation $\mathbf{h}_{\text{main}} \in \mathbb{R}^D$:

$$\mathbf{h}_{\text{main}} = \text{MLP}_{\text{int}}(\text{vec}(\mathbf{P})) \quad (14)$$

where $\text{vec}(\cdot)$ denotes the vectorization operation. The integration MLP maps the flattened $3D$ -dimensional representation back to the common feature space.

The main fusion path adaptively aggregates modality representations but does not directly parameterize their pairwise products. EPIRC supplies such terms in a parallel additive path. Each mixer output is first projected into a rank- R space:

$$\mathbf{p}^m = \mathbf{W}_x^m \mathbf{H}^m + \mathbf{b}_x^m \quad (15)$$

where $\mathbf{p}^m \in \mathbb{R}^R$ is the projected representation of modality m . Element-wise multiplication forms the three active pairwise interactions:

$$\begin{aligned} \mathbf{r}_{AV} &= \mathbf{p}^A \odot \mathbf{p}^V \\ \mathbf{r}_{TV} &= \mathbf{p}^T \odot \mathbf{p}^V \\ \mathbf{r}_{TA} &= \mathbf{p}^T \odot \mathbf{p}^A \end{aligned} \quad (16)$$

The three products are concatenated with a fixed zero slot of width R , matching the implemented $4R$ -input output projection:

$$\begin{aligned} \mathbf{r}_{\text{cat}} &= [\mathbf{r}_{AV}; \mathbf{r}_{TV}; \mathbf{r}_{TA}; \mathbf{0}_R] \\ \mathbf{r} &= \mathbf{W}_o \mathbf{r}_{\text{cat}} + \mathbf{b}_o \in \mathbb{R}^D \end{aligned} \quad (17)$$

Here $\mathbf{0}_R$ is fixed to zero, so no three-modality product enters the correction. The output layer $\mathbf{W}_o \in \mathbb{R}^{D \times 4R}$ and $\mathbf{b}_o \in \mathbb{R}^D$ are initialized to zero. Thus EPIRC initially leaves the main fusion output unchanged and can learn an additive pairwise term during training. ‘‘Correction’’ describes this residual design and does not imply that the branch has been shown to fix particular prediction errors.

With $\mathbf{h}_{\text{main}}$ and $\mathbf{r}$, the final fused multimodal representation $\mathbf{h} \in \mathbb{R}^D$ can be obtained as follows:

$$\mathbf{h} = \mathbf{h}_{\text{main}} + \mathbf{r} \quad (18)$$

Emotion Prediction and Learning Objective

To predict the emotion of the current utterance, the implementation applies three linear heads to the same fused representation and averages their logits. The average is algebraically an affine map, written with effective parameters $\mathbf{W}_c$ and $\mathbf{b}_c$:

$$\mathbf{z} = \mathbf{W}_c \mathbf{h} + \mathbf{b}_c \quad (19)$$

where $\mathbf{z} \in \mathbb{R}^C$ contains the averaged logits. The three heads remain separately parameterized during training. To optimize our model, we deploy PolyLoss (Leng et al. 2022):

$$\mathcal{L}_{\text{main}} = \frac{1}{B} \sum_{i=1}^B \omega_i \ell_{\text{poly}}(\mathbf{z}_i, y_i; \mathbf{w}) \quad (20)$$

Here, ω_i denotes the focal modulation coefficient and B is the number of samples. $\mathbf{z}_i$ is the fused multimodal emotion

logits of sample i in Eq. (19). ℓ_{poly} is per-sample Poly loss, which can be defined as follows:

$$\ell_{\text{poly}}(\mathbf{z}, y; w_y) = -w_y \log p_y + \alpha_p (1 - p_y)^{\gamma_p + 1} \quad (21)$$

where $p_y = \text{softmax}(\mathbf{z})_y$ is the probability of target class y , w_y is the class weight. α_p and γ_p control the Poly term.

Beyond the $\mathcal{L}_{\text{main}}$, we also deploy an auxiliary loss $\mathcal{L}_{\text{aux}}$ to help model learning. We add three auxiliary classifiers directly on the post-feature-gating vectors $\mathbf{u}^T$, $\mathbf{u}^A$, and $\mathbf{u}^V$, before adaptive context construction. For sample i and modality m , the corresponding auxiliary logits are

$$\mathbf{z}_{\text{aux},i}^m = \text{MLP}_{\text{aux}}^m(\mathbf{u}_i^m) \quad (22)$$

where $\text{MLP}_{\text{aux}}^m$ is a modality-specific two-layer classifier with GELU and dropout, and $\mathbf{z}_{\text{aux},i}^m \in \mathbb{R}^C$ contains its unimodal emotion logits. These classifiers are used only during training to supervise the output of feature gating, encouraging the gated representations to retain modality-specific discriminative information before cross-modal interaction. Thus, the auxiliary loss $\mathcal{L}_{\text{aux}}$ can be constructed by aggregating the unimodal Poly loss of each modality:

$$\mathcal{L}_{\text{aux}} = \sum_{m \in \{T, A, V\}} \lambda_m \frac{1}{B} \sum_{i=1}^B \ell_{\text{poly}}(\mathbf{z}_{\text{aux},i}^m, y_i; \mathbf{w}_m) \quad (23)$$

Therefore, the total learning objective is defined as the weighted sum of $\mathcal{L}_{\text{main}}$ and $\mathcal{L}_{\text{aux}}$:

$$\mathcal{L} = \lambda_{\text{main}} \mathcal{L}_{\text{main}} + \mathcal{L}_{\text{aux}} \quad (24)$$

Experiments

Experiment Settings

Datasets. We evaluate MM-Mixer on two widely used MERC benchmarks: IEMOCAP (Busso et al. 2008) and MELD (Poria et al. 2019). IEMOCAP contains approximately 12 hours of dyadic conversations with synchronized transcriptions, speech, and visual behavior. There are six classes of data in IEMOCAP, including happiness, sadness, neutral, anger, excitement, and frustration. MELD is derived from the Friends TV series and contains multi-party conversations annotated at the utterance level with seven emotion categories. It has more than 1400 dialogues and 13000 utterances with seven emotion annotations: anger, disgust, sadness, joy, neutral, surprise, and fear. We follow the standard train, validation, and test splits and classification settings for both datasets.

Baselines. We compare MM-Mixer with representative methods from three mainstream MERC paradigms. First, DialogueRNN (Majumder et al. 2019) is a sequence-based model that tracks dialogue context and speaker states with recurrent networks. Second, MMGCN (Hu et al. 2021), MM-DFN (Hu et al. 2022), and M3Net (Chen et al. 2023) are graph-based methods that model structural dependencies among utterances and modalities. Third, Ada2I (Nguyen et al. 2024), SDT (Ma et al. 2024), and CSS (Lyu et al. 2026) employ attention-based or specialized fusion mechanisms to exchange and integrate multimodal information. Together,

Model	Hap.	Sad.	Neu.	Ang.	Exc.	Fru.	ACC	WF1
DialogueRNN	37.25±4.53	70.94±0.80	53.94±1.12	60.40±0.95	64.05±1.44	59.05±0.47	58.68±0.23	58.77±0.12
MMGCN	36.07±3.02	68.65±0.39	50.63±1.79	62.65±0.55	65.48±3.88	60.70±1.11	59.31±0.96	58.86±0.87
MM-DFN	40.61±1.12	75.33±0.30	59.06±2.71	64.92±0.85	70.51±1.87	62.81±1.33	63.60±0.76	63.53±0.68
M3Net	41.71±1.79	72.92±3.11	62.33±0.91	64.53±1.06	75.62±0.93	57.72±2.25	63.96±0.37	63.70±0.30
Ada2I	48.90±5.40	79.62±0.85	60.37±3.04	61.53±3.29	72.26±1.47	64.42±1.24	65.70±1.63	65.52±1.70
SDT	62.27 ±1.41	78.32±0.77	71.46±0.48	67.90±1.01	79.21 ±0.96	65.04±1.36	71.19±0.38	71.23±0.48
CSS	<u>61.58</u> ±2.51	<u>79.74</u> ±0.67	72.27 ±0.48	<u>68.70</u> ±0.48	<u>77.22</u> ±0.31	<u>66.00</u> ±0.56	<u>71.51</u> ±0.15	<u>71.51</u> ±0.23
MM-Mixer	57.72±0.22	84.14 ±0.22	<u>71.81</u> ±0.36	71.59 ±1.06	74.04±1.12	69.15 ±0.38	72.07 ±0.15	72.19 ±0.12

Table 1: Performance comparison on the IEMOCAP dataset. All entries are reported as mean $\pm$ standard deviation in percentages. The best and second-best means are shown in bold and underlined, respectively.

Model	Neu.	Sur.	Fea.	Sad.	Joy	Dis.	Ang.	ACC	WF1
DialogueRNN	76.00±0.81	48.49±0.88	0.00±0.00	24.02±2.41	51.51±0.70	0.00±0.00	45.12±1.26	59.63±0.75	57.60±0.21
MMGCN	76.52±0.27	44.64±1.76	0.00±0.00	17.32±6.40	53.53±0.90	0.00±0.00	44.44±0.69	60.56±1.19	57.47±0.16
MM-DFN	74.75±2.19	47.76±2.66	0.94±1.33	17.88±5.49	51.73±1.72	0.00±0.00	42.18±2.43	60.29±0.39	57.95±0.24
M3Net	79.43±0.11	57.70±0.26	<u>21.79</u> ±3.65	37.45±1.39	63.89±0.87	23.67±4.27	53.68±0.96	66.78±0.17	65.39±0.22
Ada2I	77.29±0.40	49.46±1.45	0.00±0.00	25.56±3.07	56.19±0.84	0.00±0.00	48.82±0.56	62.59±0.59	59.66±0.29
SDT	79.85±0.16	<u>58.13</u> ±0.25	17.20±1.18	<u>42.64</u> ±1.92	63.95±0.20	28.34±2.01	53.63±0.59	67.06±0.44	66.09±0.39
CSS	<u>80.04</u> ±0.02	57.90±0.21	19.28±1.97	41.71±0.77	<u>64.10</u> ±0.34	<u>29.97</u> ±0.74	<u>54.36</u> ±0.46	<u>67.25</u> ±0.11	<u>66.28</u> ±0.13
MM-Mixer	80.70 ±0.16	59.79 ±0.45	29.57 ±1.94	42.87 ±0.59	65.34 ±0.59	34.93 ±3.86	55.94 ±0.55	68.36 ±0.25	67.62 ±0.23

Table 2: Performance comparison on the MELD dataset. All entries are reported as mean $\pm$ standard deviation in percentages. The best and second-best means are shown in bold and underlined, respectively.

Model	IEMOCAP	MELD
DialogueRNN	8.88M	2.67M
MMGCN	2.70M	2.55M
MM-DFN	2.06M	2.88M
M3Net	6.55M	14.11M
Ada2I	35.67M	18.56M
SDT	79.69M	68.95M
CSS	50.45M	53.40M
MM-Mixer	6.36M	6.11M

Table 3: Trainable downstream parameters in millions. Pre-trained feature extractors are excluded.

these baselines cover recurrent context modeling, graph reasoning, and multimodal attention and fusion. We retain each baseline’s released data, features, and hyperparameters. This preserves its reported setup but does not make the rows a controlled architecture-only comparison.

Evaluation metrics. We report per-class F1, accuracy (ACC), and weighted F1 (WF1). ACC and WF1 serve as the main overall metrics. Evaluation is conducted under a history-only evaluation protocol in which, when predicting the emotion of utterance u_t , a model can only take u_t and the dialogue history $\{u_1, \dots, u_{t-1}\}$ as input. We use the full dialogue history, up to 109 preceding utterances on IEMOCAP and 32 on MELD.

Implementation Details. For each utterance, we use RoBERTa-large (Liu et al. 2019), wav2vec 2.0 (Baevski et al. 2020), and DenseFace (Zhao et al. 2018) to obtain the textual, acoustic, and visual representations, respectively. Their dimensions are 1024, 1024, and 342. All modality features are projected to a common width of $D = 256$. The Modality-centric Cross-modal Alignment and learnable-query pooling use eight attention heads. The Axis-wise Multimodal Mixer

constructs $S = 6$ learned elements for each modality. Its hidden-feature MLP has a hidden width of 1536. The Explicit Pairwise Interaction Residual Correction uses an interaction rank of $R = 32$. We optimize MM-Mixer with AdamW using base learning rates of 3×10^{-5} on IEMOCAP and 4.17×10^{-5} on MELD. The weight decay is 2×10^{-4} and 9.98×10^{-5} , respectively. We use a batch size of 32 and train for up to 100 epochs on IEMOCAP and 50 epochs on MELD. The dropout rate is set to 0.2 on both datasets. The Poly-loss parameters are $\alpha_p = 1.951$ and $\gamma_p = 1.402$, with focal exponent $\gamma_f = 2.5$. On IEMOCAP, the coefficients $(\lambda_{\text{main}}, \lambda_T, \lambda_A, \lambda_V)$ are (0.45, 0.33, 0.11, 0.11). On MELD, the fused and three unimodal objectives are equally weighted. All experiments are run on an NVIDIA TESLA V100 with three seeds [0, 1, 42], and we report their average performance and standard deviation.

Main Results

Performance comparison. Table 1 reports the highest mean ACC (72.07%) and WF1 (72.19%) for MM-Mixer among the listed IEMOCAP methods. These values exceed the CSS row by 0.56% and 0.68% in ACC and WF1, respectively. The reported per-class F1 is higher than CSS for sadness by 4.40%, anger by 2.89%, and frustration by 3.15%. MM-Mixer has the highest listed F1 for these three categories, while other methods lead on happiness, neutral, and excitement. The class-wise pattern varies, and the comparison does not identify which MM-Mixer component accounts for any particular difference.

Table 2 reports the highest mean overall ACC (68.36%) and WF1 (67.62%) for MM-Mixer among the listed MELD methods, exceeding CSS by 1.11% and 1.34%, respectively. The MM-Mixer row also has the highest mean F1 in all seven listed categories. The reported gaps over CSS are largest for

Model Variant	IEMOCAP		MELD	
	WF1	ACC	WF1	ACC
Full MM-Mixer	72.16	72.03	68.01	68.77
w/o AG	71.68	71.53	67.36	68.08
w/o MCA	71.13	70.98	67.67	68.35
w/o AMM	71.44	71.29	67.05	67.60
w/o EPIRC	71.64	71.47	67.10	67.82
w/o Feature Gating	71.90	71.78	67.67	68.44
w/o Sequence Mixing	71.64	71.47	67.78	68.58
w/o Modality Mixing	71.80	71.66	67.77	68.54
w/o Feature Mixing	71.81	71.66	67.68	68.40
One Mixer Block	71.19	71.04	67.80	68.51
w/o Auxiliary Loss	71.06	70.92	67.47	68.35

Table 4: Component ablation results on IEMOCAP and MELD. AG denotes Adaptive Gating. MCA denotes Modality-centric Cross-modal Alignment. AMM denotes Axis-wise Multimodal Mixer. EPIRC denotes Explicit Pair-wise Interaction Residual Correction.

Modalities	IEMOCAP		MELD	
	WF1	ACC	WF1	ACC
T	67.98	67.90	66.71	67.43
A	54.10	54.41	48.38	52.53
V	33.66	34.44	32.60	41.92
T+A	70.29	70.24	66.81	67.58
T+V	68.76	68.64	66.37	67.01
A+V	56.55	56.62	48.64	52.68
T+A+V	72.16	72.03	68.01	68.77

Table 5: Modality ablation results on IEMOCAP and MELD. All values are percentages.

fear (10.29%) and disgust (4.96%). Nevertheless, the absolute F1 of these minority emotions remains below that of the majority emotions, and the per-class table alone cannot attribute their differences to axis-wise mixing or EPIRC.

These rankings describe the reported rows. The released Full artifacts select checkpoints by test WF1 and a three-seed subset by test performance, so they do not constitute an unbiased held-out estimate. The seed list and MELD overall values in this draft also require reconciliation with those artifacts before the comparison can support a stronger claim of general superiority.

Model efficiency. Table 3 reports 6.36M and 6.11M trainable downstream parameters for MM-Mixer on IEMOCAP and MELD, respectively, compared with 79.69M and 68.95M for SDT and 50.45M and 53.40M for CSS. Under this parameter-count convention, the reductions relative to SDT and CSS are 92.0% and 87.4% on IEMOCAP and 91.1% and 88.6% on MELD. Several graph-based baselines are smaller. The count excludes the pre-trained feature extractors and does not measure end-to-end memory use, inference time, or training cost. These draft efficiency measurements have not yet been re-audited against the released corrected Full configuration.

Ablation Study

Component ablation. Table 4 reports aggregate component variants for IEMOCAP and MELD. Each listed variant has

lower WF1 than the Full row within that table. Its Full row differs from the Full values in the main-result tables, and the released corrected Full record does not include matched variant logs or per-seed ablation results. Accordingly, the differences below are descriptive within-table contrasts; they do not establish statistically reliable, seed-stable, independent, or complementary component effects.

On IEMOCAP, the table reports WF1 reductions of 1.10% without auxiliary loss and 1.03% without MCA. Using one mixer block instead of two is lower by 0.97%. Removing latent-sequence, modality, or hidden-feature mixing is lower by 0.52%, 0.36%, and 0.35%, respectively. Removing the complete AMM is lower by 0.72%, and removing EPIRC is lower by 0.52%. These rows indicate where the reported aggregate values differ; they do not show which interaction mechanism explains the main table’s class-wise pattern.

On MELD, removing AMM or EPIRC is lower by 0.96% or 0.91% WF1 in the ablation table. Removing latent-sequence, modality, or hidden-feature mixing is lower by 0.23%, 0.24%, or 0.33%, respectively, and using one mixer block is lower by 0.21%. Removing Feature Gating is lower by 0.26% on IEMOCAP and 0.34% on MELD; removing Adaptive Gating is lower by 0.48% and 0.65%, respectively. The MCA and auxiliary-loss variants also have lower WF1 than the table’s Full row. Matched per-seed reruns are needed to determine whether any of these differences persist for the corrected Full configuration.

Modality contribution. In Table 5, text is the strongest unimodal input on both datasets, whereas audio and vision alone have lower WF1. Adding audio to text corresponds to 2.31% higher WF1 on IEMOCAP and 0.10% on MELD. The T+A+V row is highest in this table on both datasets, exceeding the best bimodal row by 1.87% and 1.20% WF1, respectively. This is consistent with useful additional modalities in the reported configurations, but the aggregate table does not establish stable gains across seeds or identify which cues the model uses.

Conclusion

MM-Mixer organizes projected utterance-level modality vectors along learned-coordinate, modality, and feature axes. Adaptive Gating and Modality-centric Cross-modal Alignment prepare the three inputs, while EPIRC adds an explicit low-rank pairwise residual to query-based fusion. The reported IEMOCAP and MELD tables show strong overall scores and compact downstream parameter counts among the listed methods. The evidence is limited to pre-extracted features on two benchmarks, and text remains the strongest single modality. In addition, the aggregate ablations lack matched per-seed provenance for the corrected Full configuration, so they do not establish stable effects or independent contributions of individual modules. Future work should reconcile the evaluation records, use validation-selected checkpoints for an unbiased held-out comparison, and examine minority emotions and information lost during utterance-level feature extraction.

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